

Research Proposal:Multimodal Fusion for Sentiment and Emotion Extraction: A Study of Robustness Across Australian and UK English Dialects

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Abstract—

In order to overcome the shortcomings of current systems that depend on either clean text or controlled acoustic conditions, this thesis suggests a novel multimodal fusion pipeline for sentiment and emotion extraction from speech. In order to generate sentiment and emotion labels, the pipeline fuses Voxtral-based transcription and internal acoustic feature extraction with an ASR-aware fine-tuned XLM-RoBERTa text classifier via a learned fusion layer. One important contribution is the ASR-aware fine-tuning strategy, which makes the text classifier resilient to the transcription noise caused by accented speech by training it on Voxtral-transcribed text instead of clean gold-standard text. Three datasets are used to assess the pipeline: MELD, CMU-MOSI, and RAVDESS. Common Voice en-AU and the English Dialects dataset to stress-test dialect robustness across regional dialects of Australian and British English. The effect of model scale on downstream extraction quality is further examined in a benchmarking study that contrasts Voxtral Mini and Voxtral Small.

I. INTRODUCTION

The extraction of sentiment and emotions from spoken language is one of the most challenging problems in natural language processing and affective computing. Although significant progress has been made in speech and sentiment analysis independently, the intersection of these fields, especially across various accents and dialects, remains largely unexplored. The foundational challenge is in the fact that speech carries meaning through two distinct channels simultaneously, the words spoken (lexical content) and the way they are spoken (acoustic properties such as tone, pitch, and rhythm). Current systems typically use one of these channels, resulting in pipelines that are fragile when deployed in real-world conditions.

The emergence of large-scale pre-trained models has transformed both the speech processing and sentiment analysis fields [1] [2] [3], combined with the Transformers-based architectures [4] and multilingual pre-training [5], have made the high-quality transcription and cross-lingual text understanding widely accessible. Most recently, Voxtral [6] introduced open-weight multimodal audio models achieving the state-of-the-art

transcription performance, internally built on a Whisper Large-v3 audio encoder, making it uniquely suited for both transcription and acoustic feature extraction. Converting speech to text before applying emotion classifiers can improve over purely acoustic approaches, establishing the speech-to-text modality conversion as a promising but unexplored direction [7].

Despite these advances, a critical gap remains. No existing work has investigated a pipeline that combines Voxtral-based transcription with internal acoustic embedding extraction and ASR-aware text classification, specifically trained on robust transcription noise introduced by the accented speech. Australian English and UK regional dialects such as Southern, Northern, Midlands, Welsh and Scottish English represent a significant and underserved population whose accents introduce systematic errors that degrade standard sentiment and emotion extraction pipelines.

This thesis addresses this gap by proposing a novel multimodal fusion pipeline for sentiment and emotion extraction from speech. The pipeline leverages Voxtral [6] for both transcription and acoustic feature extraction, and fine-tunes XLM-RoBERTa [5] on ASR-output text to build robustness to transcription noise. The methodology is evaluated across three standard datasets, MELD [8], CMU-MOSI [9], and RAVDESS [10], and stress-tested for dialect robustness on Common Voice en-AU [11] and the English Dialects dataset [12]. Two Voxtral model sizes are benchmarked to assess the impact of model scale on downstream extraction quality.

The thesis makes the following contributions:

- 1) A novel multimodal fusion pipeline combining both Voxtral acoustic embeddings and ASR-aware XLM-RoBERTa text classification for sentiment and emotion extraction.
- 2) An ASR-aware finetuning strategy that trains the text classifier on Voxtral-transcribed text rather than clean text, improving robustness to dialect transcription noise inspired by [7].
- 3) A systematic dialect robustness evaluation across Australian English and UK regional dialects.
- 4) A benchmarking study comparing Voxtral Mini and Voxtral Small as the ASR backbone.

II. RELATED WORK

A. Sequence Modeling Foundations

The foundations of modern speech and language processing rest on three landmark contributions. Graves et al. [13] introduced Connectionist Temporal Classification (CTC), a training objective that enabled recurrent neural networks to label unsegmented sequence data without requiring explicit alignment between input audio and output labels. This was the first practical framework for end-to-end speech recognition and established the sequence modelling principles that underpin all subsequent ASR architectures used in this thesis. Vaswani et al. [4] introduced the Transformer architecture, replacing recurrence entirely with self-attention mechanisms, enabling highly parallelised training on long sequences and achieving superior performance across a wide range of sequence modelling tasks. The Transformer forms the architectural backbone of every model used in this thesis - Voxtral, Whisper, and XLM-RoBERTa are all Transformer-based. Devlin et al. [14] built on the Transformer with BERT, demonstrating that bidirectional pre-training on large unlabelled text corpora produces contextual representations that transfer effectively to downstream tasks including sentiment and emotion classification. BERT established the pre-training and fine-tuning paradigm that directly motivates the XLM-RoBERTa fine-tuning strategy proposed in this thesis.

B. Self-Supervised Speech Representations

The wav2vec family of models established self-supervised learning as the dominant paradigm for speech representation learning and directly influenced the acoustic encoder architecture used in Voxtral. Schneider et al. [1] demonstrated that training a convolutional network to predict future audio frames from past context yields representations that substantially improve ASR performance when used for downstream fine-tuning, removing the dependency on large labelled speech corpora. Baevski et al. [2] extended this with vq-wav2vec, introducing discrete quantised speech representations learned without supervision, enabling the application of BERT-style pre-training directly to speech for the first time. This line of work culminated in wav2vec 2.0 [3], which jointly learns quantised speech representations and a context network through a contrastive pre-training objective, achieving near-human ASR performance with as little as ten minutes of labelled data. These models are directly relevant to this thesis as they establish the theoretical basis for extracting rich acoustic representations from pre-trained speech encoders - the same principle applied when extracting internal audio embeddings from Voxtral's Whisper-based encoder for the acoustic branch of the proposed pipeline.

C. Multilingual Speech Transcription

The development of large-scale multilingual speech transcription models provides the ASR backbone for the proposed pipeline. Radford et al. [15] introduced Whisper, trained on 680,000 hours of multilingual audio using weak supervision, demonstrating that scale and diverse training data produce

models robust to a wide range of acoustic conditions, languages, and speaker variations including accented speech. Whisper is particularly relevant as it forms the internal audio encoder of Voxtral. Bain et al. [16] extended Whisper with WhisperX, introducing phoneme-based forced alignment for accurate word-level timestamps, improving the utility of Whisper transcripts for downstream processing tasks such as sentiment and emotion classification. Barrault et al. [17] introduced SeamlessM4T, extending multilingual speech understanding to near-universal language coverage across nearly 100 languages, demonstrating the feasibility of a single model handling diverse linguistic inputs. Most critically for this thesis, Mistral AI introduced Voxtral [6], a pair of open-weights multimodal audio chat models achieving state-of-the-art transcription performance across English and multilingual benchmarks. Voxtral is built on a Whisper Large-v3 audio encoder internally, meaning its acoustic representations inherit Whisper's robustness properties while benefiting from additional adapter and language decoder training. This makes Voxtral uniquely suited as the single backbone for a pipeline requiring both high-quality transcription and rich acoustic embeddings.

D. Sentiment and Emotion Datasets

Three datasets form the primary training and evaluation foundation for the proposed pipeline. Poria et al. [8] introduced MELD, a multimodal multi-party dataset containing approximately 13,000 utterances from the Friends TV series, annotated with seven emotion categories and three sentiment classes across audio, visual, and textual modalities. MELD is particularly well suited to this thesis due to its conversational and naturalistic nature, and its provision of both emotion and sentiment labels from the same dataset. Zadeh et al. [9] introduced CMU-MOSI, a multimodal corpus of sentiment intensity annotations for opinion video segments, providing continuous sentiment scores that enable both binary and fine-grained sentiment evaluation and serving as a standard benchmark for multimodal sentiment analysis research. Demszky et al. [18] contributed GoEmotions, the largest manually annotated fine-grained emotion dataset comprising 58,000 English Reddit comments labelled across 27 emotion categories, providing a rich resource for text-based emotion classification that complements the audio-grounded datasets. Together these three datasets cover categorical, continuous, and fine-grained annotation approaches, providing comprehensive coverage of the sentiment and emotion extraction tasks addressed in this thesis.

E. Multilingual Text Representations

The text branch of the proposed pipeline relies on cross-lingual pre-trained representations. Conneau et al. [5] introduced XLM-RoBERTa, a cross-lingual language model pre-trained on 2.5 terabytes of filtered CommonCrawl data across 100 languages. XLM-RoBERTa demonstrated that massively multilingual pre-training produces representations that transfer effectively across languages in both zero-shot and fine-tuned

settings, significantly outperforming multilingual BERT on cross-lingual benchmarks. It is adopted in this thesis as the text classification backbone and fine-tuned specifically on Voxtral-transcribed text rather than clean text - a novel adaptation not explored in prior work. Barbieri et al. [19] fine-tuned XLM-RoBERTa specifically for sentiment analysis on multilingual Twitter data, producing the CardiffNLP XLM-T model that represents the strongest existing off-the-shelf multilingual sentiment baseline. This model serves as a direct comparison point in the benchmarking experiments, establishing the performance ceiling of existing sentiment-fine-tuned multilingual models against the ASR-aware fine-tuning strategy proposed in this thesis.

F. Sentiment and Emotion Recognition

Schuller [20] provided a comprehensive two-decade survey of speech emotion recognition, identifying the central challenges of speaker variability, acted versus naturalistic data, class imbalance, and the generalisation gap between laboratory conditions and real-world deployment. This survey contextualises the limitations of purely acoustic approaches and motivates the multimodal and text-based alternatives explored in this thesis. Critically, Schuller highlights that no single modality reliably captures the full range of emotional expression - a finding that directly motivates the fusion approach proposed here. Taghavi et al. [7] directly addressed the modality question by demonstrating that converting speech to text before applying emotion classifiers improves performance on the MELD dataset, establishing speech-to-text modality conversion as a viable and underexplored direction for speech emotion recognition. This work is the most direct conceptual predecessor to the text branch of the proposed pipeline. This thesis extends it by retaining acoustic features alongside the text branch and fusing both modalities through a learned fusion layer, directly addressing the limitation that pure text-based approaches discard prosodic emotion signal that is particularly important for accented speech.

G. Multimodal Fusion

Zadeh et al. [21] introduced the Tensor Fusion Network, which models both intra-modal and inter-modal dynamics through a three-fold Cartesian product of modality embeddings, producing a joint representation for multimodal sentiment analysis. This work established the theoretical and empirical case for multimodal fusion in affective computing, demonstrating that combining acoustic, visual, and lexical signals consistently outperforms any single modality across standard benchmarks. The fusion approach proposed in this thesis draws directly on this foundation, adopting concatenation-based fusion as the baseline approach for combining text and acoustic embeddings, with the Tensor Fusion Network serving as the theoretical reference point justifying the multimodal design. The ablation study directly tests whether the fusion architecture contributes meaningfully over single-modality baselines, mirroring the experimental design established by Zadeh et al.

III. RESEARCH METHODOLOGY

A. Overview

The multimodal fusion pipeline for sentiment and emotion extraction from speech takes raw audio as input and produces sentiment and emotion labels as output by combining into parallel branches, a text branch and an acoustic branch, fused together through a learned fusion layer. The pipeline is evaluated on three standard datasets and stress-tested for robustness across Australian and UK English dialects.

B. Pipeline Architecture

Text Branch A transcript is created by passing raw audio through Voxtral [6]. The transcript is sent to XLM-RoBERTa [5], which produces a text representation for classification after being specifically tuned on Voxtral-transcribed text rather than clean text.

Acoustic Branch Voxtral’s internal Whisper Large-v3 [15] audio encoder is used to process the same audio. Internal acoustic embeddings, which capture prosodic characteristics like tone, pitch, and rhythm regardless of the words spoken, are taken straight out of the encoder. **Fusion Layer** An emotion label and a sentiment label are simultaneously produced by concatenating the text and acoustic representations and passing them through a classification head.

While the Tensor Fusion Network [21] represents a more expressive fusion approach, concatenation-based fusion is adopted as the baseline for computational efficiency and interpretability, with more complex fusion explored as a potential extension if time permits.

C. ASR-Aware Fine-Tuning

When applied to ASR output, standard text classifiers are brittle because they were trained on clean text. This thesis extends the modality conversion approach of Taghavi et al. [7] by proposing an ASR-aware fine-tuning strategy. Voxtral is used to first transcribe all training audio, and the original emotion and sentiment labels are used to refine XLM-RoBERTa on these transcripts. As a result, the classifier is resistant to the particular dialectal error patterns and transcription noise that Voxtral introduces.

D. Datasets

Training and Evaluation:

- MELD 13,000 labels for sentiment, emotion, and conversational utterances
- CMU-MOSI - opinion video segments, sentiment labels
- RAVDESS - acted speech recordings, emotion labels
- GoEmotions - fine-grained emotion labels, text-based

Dialect Robustness Evaluation (inference only):

- English Dialects - Southern, Midlands, Northern, Welsh, and Scottish English
- Common Voice en-AU

E. Model Variants

Two Voxtral model sizes are benchmarked to assess the impact of model scale on downstream extraction quality [6]:

- Voxtral Mini (3B) - lightweight, fast inference
- Voxtral Small (24B) - state-of-the-art performance

F. Baselines

- 1) Text-only with clean text - XLM-RoBERTa on gold-standard transcripts
- 2) Text-only with ASR text - XLM-RoBERTa on Voxtral transcripts
- 3) Off-the-shelf sentiment - CardiffNLP XLM-T [19]
- 4) Acoustic-only - Voxtral internal embeddings without text branch
- 5) Fusion with Voxtral Mini - proposed pipeline, lightweight
- 6) Fusion with Voxtral Small - proposed pipeline, full performance

G. Ablation Study

- 1) **A1** - Remove acoustic branch: text alone is insufficient
- 2) **A2** - Remove text branch: acoustics alone is insufficient
- 3) **A3** - Clean-text vs ASR-text fine-tuning: validates ASR-aware training
- 4) **A4** - Voxtral vs Whisper transcription: validates Voxtral's contribution
- 5) **A5** - Concatenation vs sum fusion: validates fusion architecture

H. Evaluation Metrics

- **Weighted F1** - primary metric, accounts for class imbalance
- **Accuracy** - secondary metric
- **Per-class F1** - identifies hardest emotion categories
- **Per-accent breakdown** - dialect robustness analysis across AU and UK subsets

IV. PROJECT MANAGEMENT

A. Timeline

- **March 2026** - Environment setup, GPU configuration, dataset preprocessing, and basic Voxtral inference pipeline
- **April 2026** - XLM-RoBERTa fine-tuning on ASR-output text, acoustic embedding extraction, and baseline experiments
- **May 2026** - Fusion layer implementation, full pipeline training, and all experiments including ablations
- **June 2026** - Error analysis, dialect robustness evaluation on AU and UK datasets, figures and tables
- **July 2026** - Dissertation writing begins with all experimental results finalised

B. Hardware

All experiments are conducted on the `wmlg-ada` partition consisting of 4x Nvidia L40S GPUs (48GB each), providing 192GB total VRAM. This is sufficient to run Voxtral Small (24B) across two GPUs, fine-tune XLM-RoBERTa on a single GPU, and train the fusion layer with capacity to spare.

C. Risk Management

- **Dataset availability** - All datasets are publicly available and verified prior to March
- **Voxtral embedding extraction** - Internal embedding access requires non-standard forward passes; one week allocated in March for this specifically
- **Scope creep** - Fusion architecture kept simple (concatenation) as baseline; complexity only added if time allows

V. CONCLUSION

In order to extract sentiment and emotion from speech, this thesis proposes a novel multimodal fusion pipeline that combines ASR-aware XLM-RoBERTa text classification with Voxtral-based transcription and acoustic feature extraction. A benchmarking study comparing Voxtral Mini and Voxtral Small as the ASR backbone is conducted, and the pipeline is assessed on three standard datasets and stress-tested for robustness across Australian and UK English dialects. By addressing the shortcomings of current systems, the suggested multimodal fusion approach and ASR-aware fine-tuning strategy lay the groundwork for more reliable spoken language understanding across the vast population of speakers whose accents are still underserved by existing pipelines.

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